Week 10 2–6 November 2026

Subject: Computer science Unit: A1.3 Operating systems Grade: 11DP SL

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 OS functions 60 minutes	A1.3.2 Describe OS functions. Explain system integrity; memory, file and device management; scheduling; security; accounting; GUI; virtualization; networking. Apply the functions to a familiar computer-use scenario.	5 min: retrieve OS role. 15 min: model the functions. 25 min: pairs match all ten functions to examples, then annotate a download-and-save scenario. 10 min: individual exam-style description. 5 min: exit ticket. Support: glossary and matching cards. Extend: distinguish accounting/security and explain interacting functions.	A1.3.2.docx; IB guide p. 30; OS-function cards; teacher Task Manager demonstration. Homework: describe four OS functions using examples; revisit the attached notes in the shared class folder.
Lesson 2 CPU scheduling 60 minutes	A1.3.3 Compare scheduling approaches. Describe FCFS, round robin, multilevel queue and priority scheduling. Compare response, fairness, overhead and starvation; justify an approach for a workload.	5 min: retrieve scheduling purpose. 15 min: model four methods. 20 min: trace shared process cards using FCFS and round robin, then classify multilevel/priority cases. 15 min: compare and recommend for a desktop/batch system. 5 min: exit ticket. Support: queue templates and stated quantum/priority order. Extend: explain aging and quantum trade-offs.	A1.3.3.docx; IB guide p. 30; process cards; queue/timeline worksheet. Homework: compare two methods and justify one for an interactive system; review the attached notes.
Lesson 3 Polling and interrupts 60 minutes	A1.3.4 Evaluate polling and interrupt handling. Explain checking versus notification and the interrupt save–handle–restore sequence. Evaluate event frequency, overhead, power, predictability, latency and security in real scenarios.	5 min: retrieval. 10 min: demonstrate polling/interrupts. 20 min: evaluate a battery sensor, keyboard, network and real-time system against the six factors. 20 min: individual evaluation with evidence-based judgment and peer feedback. 5 min: exit ticket. Support: comparison grid and sentence stems. Extend: explain high-rate interrupt overhead and latency limits.	A1.3.4.docx; IB guide p. 30; event cards and evaluation grid. Homework: evaluate a suitable method for one embedded-system scenario; use the attached notes.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Use exit tickets to plan retrieval or reteaching. Specify arrival order, time quantum, priority convention and switching assumptions in scheduling tasks. Neither polling nor interrupts is always best; require a justified choice. Scope this week: A1.3.2–A1.3.4 (SL content); HL A1.3.5 is outside this plan.		

Week 10 2–6 November 2026

Technology Integration دمج التكنولوجيا	Demonstrate CPU/memory use in Task Manager using non-sensitive examples. Students record scheduling traces and compare polling/interrupt event flows in a shared worksheet.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: sequencing and time allocation. Physics: input/output events and power use. Citizenship: fair allocation, privacy and secure access to shared resources.
External Assessment Integration دمج التقييمات الخارجية	Paper 1 preparation: describe OS functions, compare scheduling methods and evaluate event handling. Model command terms, then use individual scenario responses and feedback on technical accuracy and justified judgments.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: compare alternatives and justify trade-offs. Communication: explain a queue trace to a partner. Self-management: use exit tickets to identify one misconception. Rotate roles and allow individual thinking before discussion.
Standards المعايير	IB Computer science guide, first assessment 2027, p. 30: A1.3.2, A1.3.3 and A1.3.4. Teacher notes: A1.3.2.docx, A1.3.3.docx and A1.3.4.docx.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	TOK: Is the fairest way of allocating a shared resource always the most efficient? Contrast arrival order, equal time slices and priority; distinguish measurable performance from value judgments.
Success Criteria معايير النجاح (Measurable)	MUST: Identify all ten OS functions and correctly describe at least four; outline four scheduling methods; distinguish polling from interrupts. SHOULD: Trace FCFS/round robin; compare two scheduling methods using two criteria; evaluate a scenario using at least three of the six factors. COULD: Justify a conditional recommendation addressing a counterargument, starvation/aging or interrupt overhead.
National Identity Domain مجالات الربط بالهوية الوطنية	Context: use hypothetical UAE school computer labs, shared printers and smart-building sensors to discuss resource allocation and event handling.
	Values: connect OS access permissions to privacy and responsible use of shared school devices; consider fairness when assigning priorities.
	Application: recommend an event-handling method for a hypothetical battery-powered building sensor, linking energy use, responsiveness and reliable operation.